

Design for class “CartScreen”

Inheritance of class “CustomerScreen”

CartScreen
- placeOrderController : PlaceOrderController - selectedItems : List<Product>
+ setBController(bController : BaseController) : void + navigateTo(screen : BaseScreenHandler) : void + setController(controller : PlaceOrderController) : void + handlePlaceOrderClick() : void + showQuantityNotSufficient(message : String) : void + displayCartItems()

Table 1. Attribute Design

#	Name	Data type	Default value	Description
1	placeOrderController	PlaceOrderController	null	Controller managing the Place Order flow.
2	selectedItems	List<Product>	empty list	The list of products currently in the cart.

Table 2. Operation Design

#	Name	Return type	Description
1	setController(controller)	void	Assign the PlaceOrderController to the screen.
2	handlePlaceOrderClick()	void	Trigger placing order when the user clicks the button.
3	showQuantityNotSufficient (message)	void	Display an error when one or more items have insufficient stock.
4	displayCartItems()	void	Load and render all selected cart items on screen.

Method Design:

1. setController

- Parameter:
 - controller: PlaceOrderController
- Exception:
- Method:
 - Assigns the controller to this screen.

2. handlePlaceOrderClick

- Parameter:
- Exception:
- Method:
 - Collect items in the cart.
 - Call `placeOrderController.requestToPlaceOrder(cart)`.
 - Navigate to the `DeliveryInfoScreen` if successful.

3. showQuantityNotSufficient

- Parameter:
 - message: String - error to display.
- Exception:
- Method:
 - Opens an error dialog or inline alert.

4. displayCartItems

- Parameter:
- Exception:
- Method:
 - Retrieve items from the cart entity.
 - Render them in UI.